

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2316

James Black Tap 100 kV
May 30, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2316. The Interconnection Customer proposes a 146.0 MW solar photovoltaic facility interconnecting at the James Black Tap 100 kV substation in the Ohio area. The request was received on May 25, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,871,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rio Bravo to Sherwood Forest 100kV
Contingency: Tanners Lake to Lake 100kV
Loading: 110.6% (Rating: 163 MVA)
DFAX: 17.7%
Pre-project loading: 94.7%
2. Facility: Lions Rd to Aes Beaver Vly Partners Beaver Vly 500kV L
Contingency: Cranberry to Beaver Vly 500kV
Loading: 101.1% (Rating: 2,652 MVA)
DFAX: 8.0%
Pre-project loading: 100.6%
3. Facility: Labarre to Port Washington Sw Yd 100kV
Contingency: Statesville Delivery No 3 to Apopka South 100kV Line
Loading: 100.9% (Rating: 119 MVA)
DFAX: 7.8%
Pre-project loading: 91.3%
4. Facility: Cranberry to Beaver Vly 500kV
Contingency: Aes Beaver Vly Partners Beaver Valley to Lions Road 500kV
Loading: 100.8% (Rating: 2,657 MVA)
DFAX: 7.9%
Pre-project loading: 100.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sherwood Forest to Rio Bravo 100kV	Rebuild line with higher-capacity conductor	\$7,985,000	72% (\$5,714,000)
Lions Rd to Aes Beaver Vly Partners Beaver Vly 500kV L	Reconductor line	\$51,144,000	2% (\$758,000)
Labarre to Port Washington Sw Yd 100kV L	Reconductor line	\$3,074,000	65% (\$2,006,000)
Cranberry to Beaver Vly 500kV	Reconductor line	\$26,441,000	2% (\$393,000)
	Total allocated to project		\$8,871,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.